Generative Modeling of Molecular Dynamics Trajectories

NeurIPS 2024

动力学任务

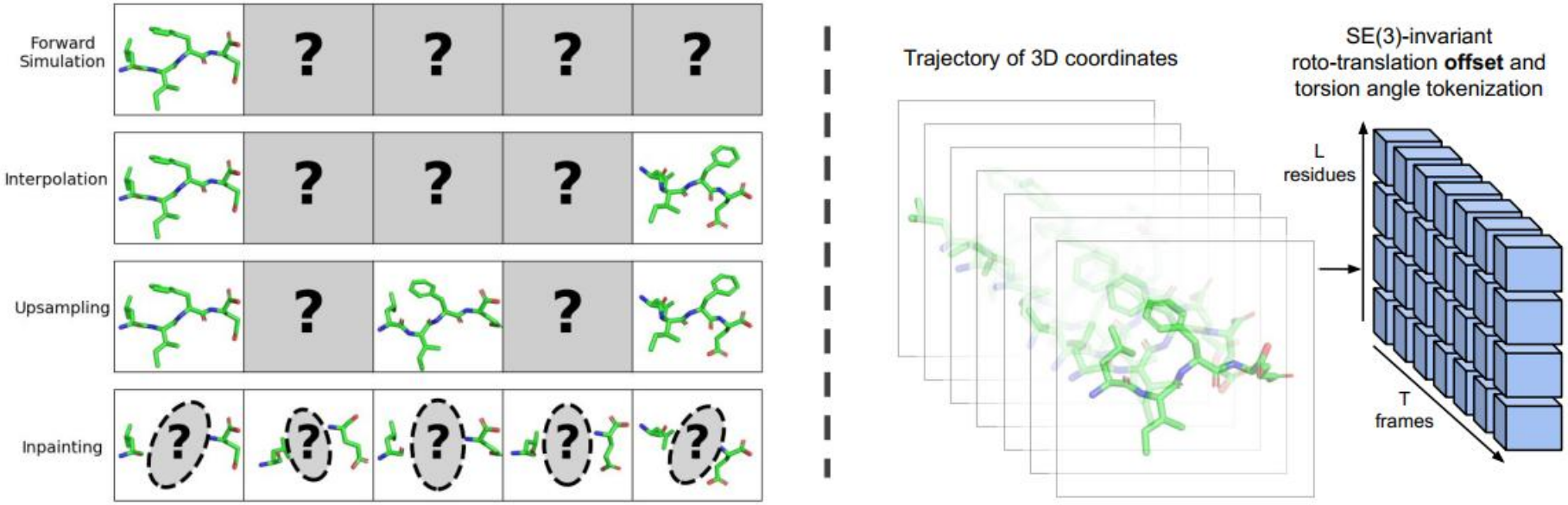
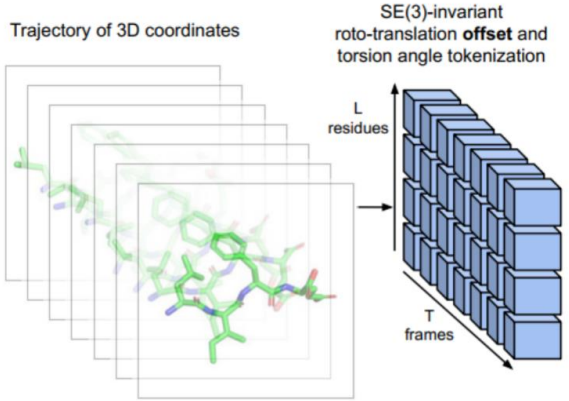


Figure 1: (*Left*) Tasks: generative modeling of MD trajectories addresses several tasks by conditioning on different parts of a trajectory. (*Right*) Method: We tokenize trajectories of T frames and L residues into an $(T \times L)$-array of SE(3)-invariant tokens encoding roto-translation offsets from *key frames* and torsion angles. Using *stochastic interpolants*, we generate arrays of such tokens from Gaussian noise.

问题定义



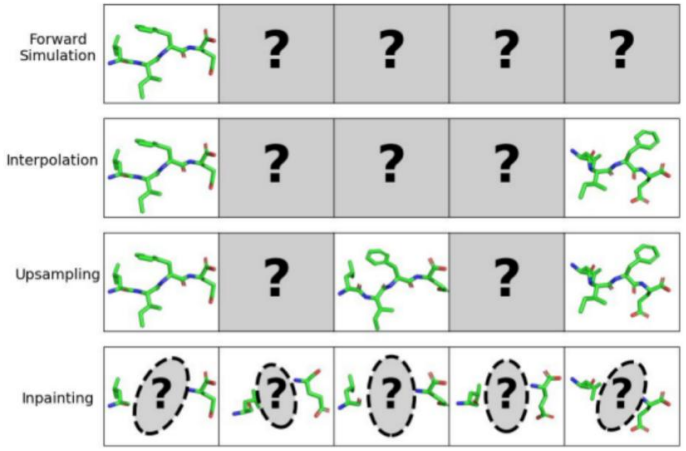
轨迹数据表示：

$$\chi_t^l = [g, \tau_1, \ldots \tau_7], \quad g \in SE(3), \tau \in \mathbb{T}, \quad \chi \in \left(\left[SE(3) \times \mathbb{T}^7 \right]^L \right)^T$$

$$\chi_t^j = \left[\phi\left([g_{t_1}^j]^{-1} g_t^j \right), \ldots, \phi\left([g_{t_K}^j]^{-1} g_t^j \right), \psi([\tau_t^j]_1), \ldots \psi([\tau_t^j]_7) \right] \subset \mathbb{R}^{7K+14}$$

学习目标：

$$v_\theta(\cdot \mid g_{t_1} \ldots g_{t_K}, A) : \mathbb{R}^{T \times L \times (7K+14)} \times [0,1] \to \mathbb{R}^{T \times L \times (7K+14)}$$



多任务：

Setting	Key frames	Generate	Conditioned on	Token dim.
Forward simulation	g_1	$g_{1\cdots T}, \boldsymbol{\tau}_{1\cdots T}$	$g_1, \boldsymbol{\tau}_1, A$	21
Interpolation	g_1, g_T	$g_{1\cdots T}, \boldsymbol{\tau}_{1\cdots T}$	$g_{1,T}, \boldsymbol{\tau}_{1,T}, A$	28
Upsampling	g_1	$g_{1\cdots T}, \boldsymbol{\tau}_{1\cdots T}$	$g_{1+\{1,2,\cdots\}M}, \boldsymbol{\tau}_{1+\{1,2,\cdots\}M}, A$	21
Inpainting	g_1, g_T	$g_{1\cdots T}, A$	$g_{1\cdots T}^{\text{known}}$	7 (+20)

前向生成任务评测

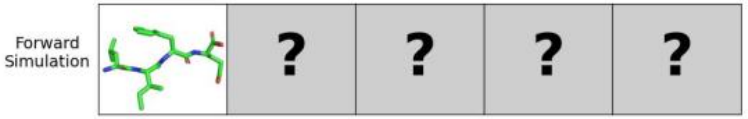
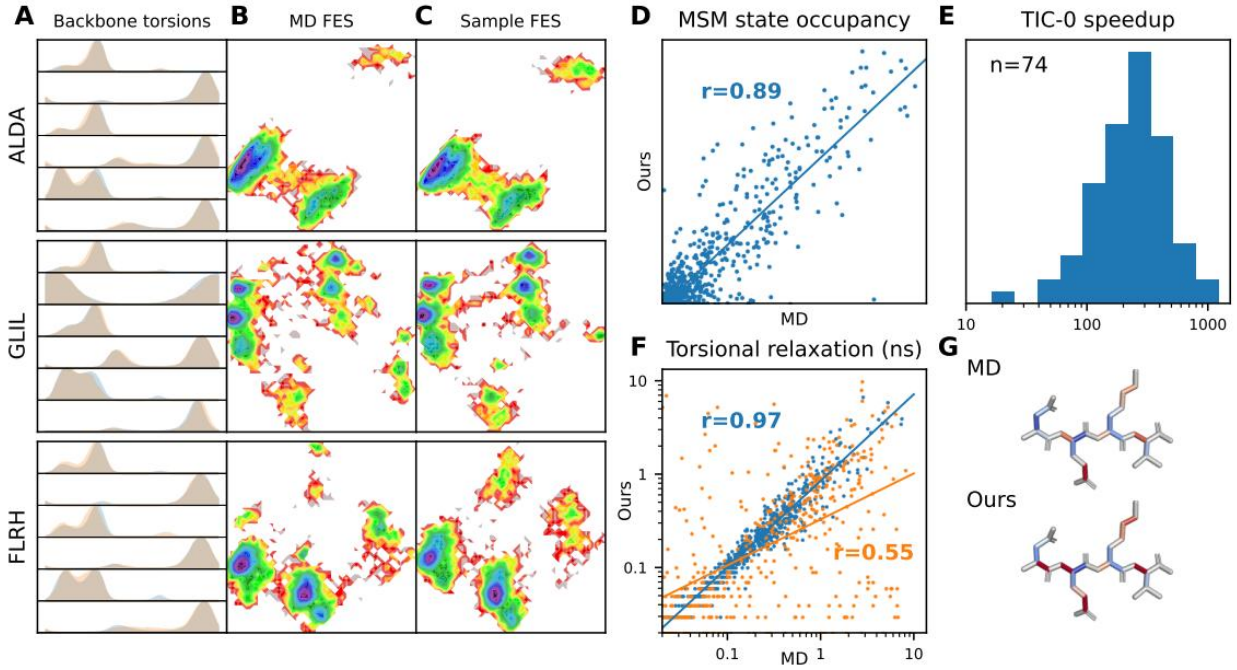


Table 2: JSD between sampled and ground-truth distributions, with replicate simulations as baselines. 100 ns represents oracle performance.

C.V.	Ours	10 ns	1 ns	100 ps	100 ns
Torsions (bb)	**.130**	.145	.212	.311	.103
Torsions (sc)	**.093**	.111	.261	.403	.055
Torsions (all)	**.109**	.125	.240	.364	.076
TICA-0	**.230**	.323	.432	.477	.201
TICA-0,1 joint	**.316**	.424	.568	.643	.268
MSM states	**.235**	.363	.493	.527	.208
Runtime	60s	1067s	107s	11s	3h



四肽数据集： 3000 training, 100 validation, 100 test

生成过程： 每次生成10ns，间隔为10ps，生成100帧，拼接到100ns

评价指标： 角度，分子慢动作降维分布，马尔可夫状态模型分布（把构象空间切分为微状态）

插值任务评测

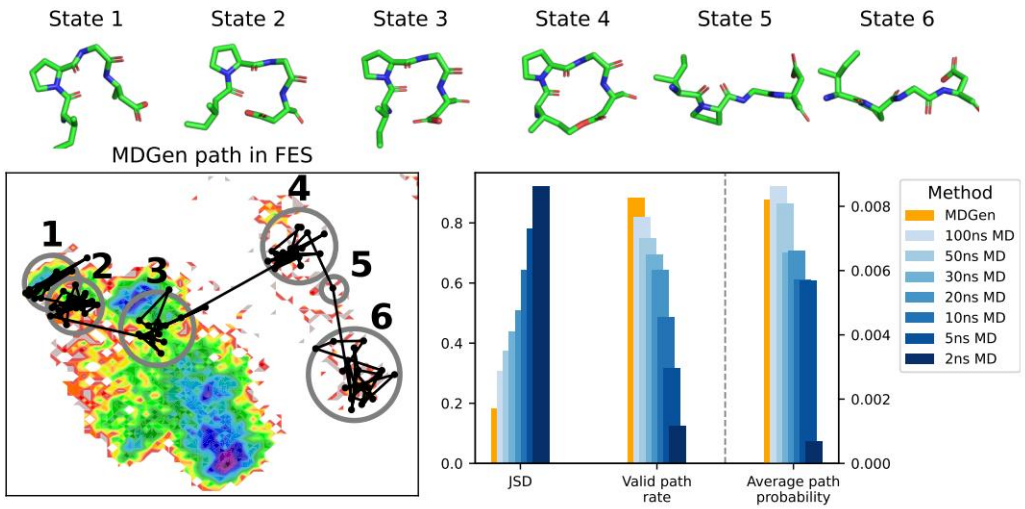


Figure 3: **Transition path sampling results.** (**Top**) Intermediate states of one of the 1-nanosecond interpolated trajectories between two metastable states for the test peptide IPGD. (**Bottom Left**) The corresponding trajectory on the 2D free energy surface of the top two TICA components (more examples in Figure 9). (**Bottom Right**) Statistics averaged over 100 test peptides and 1000 paths for each of them. Shown are JSD, fraction of drawn paths that are valid transition paths, and average path likelihood of our discretized transitions under the reference MSM compared to discrete transitions drawn from the reference MSM or alternative MSMs built from replica simulations of varying lengths.

四肽数据集：3000 training, 100 validation, 100 test

训练过程：训练的时候生成1ns，10帧

测试过程：选择most well-separated states (能垒很高)的两个点生成，采样1000个路径

评价指标为：比较根据离散状态直接的转移路径分布和MSM的状态转移矩阵之间的path likelihood

插值任务评测

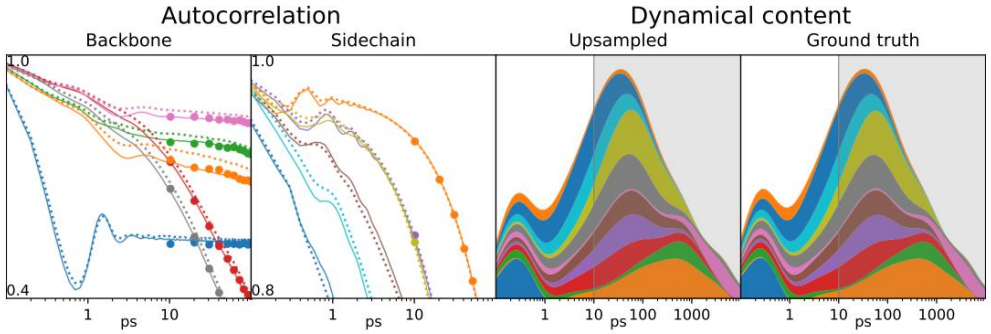


Figure 4: **Recovery of fast dynamics via trajectory upsampling** for peptide GTLM. (*Left*) Autocorrelations of each torsion angle from (—) the original 100 fs-timestep trajectory, (•) the subsampled 10 ns-timestep trajectory, and (···) the reconstructed 100 fs-timestep trajectory (all length 100 ns). (*Right*) Dynamical content as a function of timescale from the upsampled vs. ground truth trajectories, stacked for all torsion angles (same color scheme). The subsampled trajectory contains only the shaded region and our model recovers the unshaded region. Further examples in Figure 10.

Table 3: Sequence recovery for the inner two peptides when conditioning on the partial trajectory (MDGEN), the two terminal frames (DynMPNN), or a single frame (S-MPNN).

Method	High Flux	Random Path
MDGen	**52.1%**	**62.0%**
DynMPNN	17.4%	24.5%
S-MPNN	16.3%	13.5%

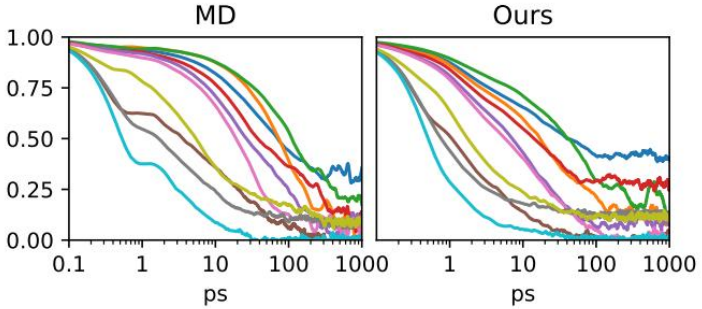


Figure 5: Autocorrelation functions of MDGEN sidechain torsion angles computed from a 10-ns MD trajectory (*left*) versus a single **100k-frame model sample** with Hyena (*right*), capturing dynamics spanning four orders of magnitude.

长时间模拟

蛋白质轨迹

Table 4: Median results on test protein ensembles ($n = 82$); evaluations from Jing et al. (2024). Runtimes are reported per sample structure or frame.

	MDGEN	AlphaFlow	MSA sub.
Pairwise RMSD $r \uparrow$	**0.48**	**0.48**	0.22
Global RMSF $r \uparrow$	0.50	**0.60**	0.29
Per-target RMSF $r \uparrow$	0.71	**0.85**	0.55
Root mean $\mathcal{W}_2$ dist. $\downarrow$	2.69	**2.61**	3.62
MD PCA $\mathcal{W}_2$ dist. $\downarrow$	1.89	**1.52**	1.88
% PC-sim > 0.5 $\uparrow$	10	**44**	21
Weak contacts $J \uparrow$	0.51	**0.62**	0.40
Exposed residue $J \uparrow$	0.29	**0.41**	0.27
Runtime (s)	**0.2**	70	4

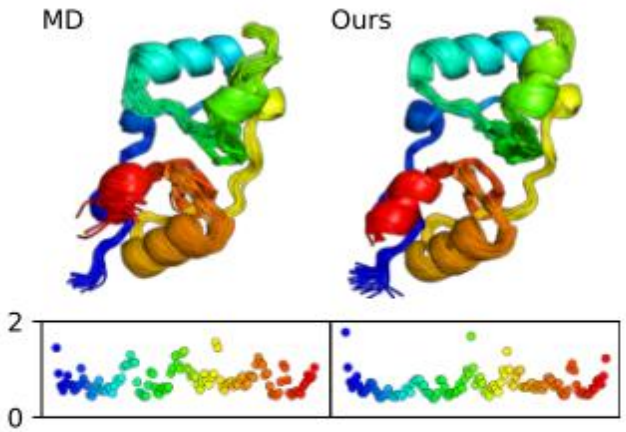


Figure 6: MD vs generated ensembles for 6uof_A, with Cα RMSFs plotted by residue index (Pearson $r = 0.74$).

蛋白质动力学数据集
ATLAS (Confrover)
1938 samples

生成250帧，步长400ps